

A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

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Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo 2^{B_M+1} has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and M — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

1 Conventions

For a valuation word $w = (b_0, \dots, b_{M-1})$, $B_N = \sum_{n < N} b_n$; the set of odd x whose first N valuations equal the prefix is a single residue class modulo 2^{B_N+1} ; “rep bits” is the bit length of its least positive representative r_N , “mod bits” is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force r_N to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

2 Translating between a word and the integer it represents

Explanation

A word does not represent a single integer but a single *residue class*: the set of odd x whose first M valuations equal $w = (b_0, \dots, b_{M-1})$ is exactly one arithmetic progression

$$x \equiv r \pmod{2^{B_M+1}}, \quad B_M = \sum_{n < M} b_n,$$

and “the integer” tabulated in this catalog is its least positive representative r . Every member $r + k 2^{B_M+1}$ ($k \geq 0$) produces the same first M letters; the word sees only $x \bmod 2^{B_M+1}$. The converse direction is trivial: iterate $x \mapsto (3x + 1)/2^{v_2(3x+1)}$ and read the exponents.

Formula

Along the prefix the orbit point has the exact affine form

$$x_n = \frac{3^n x_0 + A_n}{2^{B_n}}, \quad A_n = \sum_{j < n} 3^{n-1-j} 2^{B_j}, \quad B_n = \sum_{j < n} b_j.$$

The letter condition $v_2(3x_n + 1) = b_n$ is the congruence

$$3^{n+1}x_0 + A_{n+1} \equiv 2^{B_{n+1}} \pmod{2^{B_{n+1}+1}} \iff x_0 \equiv 3^{-(n+1)}(2^{B_{n+1}} - A_{n+1}) \pmod{2^{B_{n+1}+1}},$$

(3 is a unit in $\mathbb{Z}_2$), so each letter appends b_n new binary digits of x_0 : a Hensel-style lift. After M letters the class modulus is 2^{B_M+1} and r is the lift's value reduced to $[1, 2^{B_M+1})$.

Python

The lift as implemented in `search/frontier.py` (`realize`); r is built letter by letter:

```
def realize(bs):
    # word -> (r, B+1)
    r, Mb = 1, 1
    # class rep and its bit-modulus
    A = 0; B = 0; P3 = 1
    # running A_n, B_n, 3^n
    for b in bs:
        need = B + b; mod = 1 << (need + 1)
        c = (3*A + (1 << B)) % mod
        # A_{n+1} mod 2^{B+b+1}
        x0 = (pow(3*P3, -1, mod)
              * (((1 << need) - c) % mod)) % mod
        if need + 1 > Mb:
            r = x0; Mb = need + 1
        A = 3*A + (1 << B); P3 *= 3; B += b
    return r, Mb

def word_of(x, M):
    # integer -> word
    out = []
    for _ in range(M):
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append(b); x = t >> b
    return out
```

Commands

```
cd shadow_frontier
# word -> integer (paste the letters as one string)
python3 -c "import sys; sys.path.insert(0,'search'); \
from frontier import realize; \
w=[int(c) for c in '111411211121...']; \
r,Mb=realize(w); print(r, r.bit_length(), Mb)"
# integer -> word, plus the four-panel candidate image
python3 search/candidate_image.py --word 1,1,1,4,1,1,2,... --out viz/cand.png
python3 search/candidate_image.py --seed 687871 --steps 55 --out viz/c687871.png
```

Worked example: catalog word #1

For the word #1 above ($M = 240$, mean 1.583, $|D_3| = 0.214$): $B_M = 380$, modulus 2^{381} , and the least positive representative is the 381-bit integer

45276335476484047115725251008950986904354815513999907611
28044450220436219657540681487193488995285987515743750727311

whose first 240 Syracuse letters reproduce the word exactly (machine-checked). Its fill ratio is r 's bit length over $B_M + 1$: $381/381 = 1.000$ — and the half-word prefix already needs 184 bits of 185: the representative *grows with the prefix* (184 $\rightarrow$ 381 bits from $M = 120$ to 240) instead of stabilizing. That diagonal ride is the ghost signature of the Conventions section: the class contains no positive integer below 2^{380} . Contrast a true integer realization: the survival champion 687871 *is* the least representative of its own 50-letter class, and stays put as the prefix grows — stabilization versus diagonal is the whole test.

3 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios ≈ 1 .

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
1	1.5833	0.214	225	380	381	381	1.000
2	1.5792	0.214	225	379	379	380	0.997
3	1.5792	0.214	225	379	378	380	0.995
4	1.5833	0.239	205	380	381	381	1.000
5	1.5917	0.220	213	382	383	383	1.000
6	1.5792	0.215	225	379	379	380	0.997
7	1.5917	0.215	225	382	383	383	1.000
8	1.5792	0.216	225	379	380	380	1.000
9	1.5792	0.229	214	379	380	380	1.000
10	1.5792	0.233	225	379	380	380	1.000
11	1.5792	0.229	225	379	379	380	0.997
12	1.5792	0.216	225	379	380	380	1.000
13	1.5917	0.233	193	382	382	383	0.997
14	1.5792	0.215	224	379	380	380	1.000
15	1.5917	0.216	208	382	383	383	1.000
16	1.5833	0.214	225	380	378	381	0.992
17	1.5917	0.215	225	382	380	383	0.992
18	1.5792	0.218	225	379	380	380	1.000
19	1.5917	0.220	225	382	383	383	1.000
20	1.5792	0.260	214	379	380	380	1.000
21	1.5792	0.272	208	379	380	380	1.000
22	1.5917	0.220	214	382	383	383	1.000
23	1.5875	0.228	225	381	382	382	1.000
24	1.5875	0.246	214	381	380	382	0.995
25	1.5917	0.222	211	382	383	383	1.000
26	1.5917	0.242	189	382	383	383	1.000
27	1.5792	0.222	222	379	376	380	0.989
28	1.5917	0.214	211	382	383	383	1.000
29	1.5875	0.215	225	381	382	382	1.000
30	1.5792	0.260	217	379	375	380	0.987
31	1.5917	0.214	216	382	382	383	0.997
32	1.5833	0.239	203	380	380	381	0.997
33	1.5792	0.218	225	379	377	380	0.992
34	1.5875	0.217	225	381	380	382	0.995
35	1.5917	0.214	224	382	379	383	0.990
36	1.5917	0.216	225	382	377	383	0.984
37	1.5792	0.220	225	379	380	380	1.000
38	1.5917	0.225	225	382	383	383	1.000
39	1.5875	0.221	225	381	381	382	0.997
40	1.5792	0.218	225	379	379	380	0.997
41	1.5917	0.225	206	382	380	383	0.992
42	1.5792	0.238	212	379	377	380	0.992
43	1.5792	0.215	225	379	375	380	0.987
44	1.5917	0.215	225	382	383	383	1.000
45	1.5875	0.217	225	381	379	382	0.992
46	1.5792	0.215	219	379	378	380	0.995
47	1.5792	0.229	223	379	380	380	1.000
48	1.5792	0.215	225	379	377	380	0.992
49	1.5917	0.215	225	382	383	383	1.000
50	1.5917	0.216	225	382	382	383	0.997

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
51	1.5792	0.229	223	379	377	380	0.992
52	1.5792	0.248	206	379	380	380	1.000
53	1.5917	0.218	224	382	376	383	0.982
54	1.5792	0.215	225	379	380	380	1.000
55	1.5917	0.233	225	382	382	383	0.997
56	1.5917	0.279	194	382	383	383	1.000
57	1.5792	0.215	225	379	380	380	1.000
58	1.5792	0.218	225	379	377	380	0.992
59	1.5792	0.238	201	379	380	380	1.000
60	1.5792	0.214	225	379	380	380	1.000
61	1.5792	0.238	225	379	380	380	1.000
62	1.5833	0.230	213	380	381	381	1.000
63	1.5792	0.222	219	379	379	380	0.997
64	1.5792	0.215	219	379	379	380	0.997
65	1.5792	0.222	225	379	380	380	1.000
66	1.5917	0.216	196	382	383	383	1.000
67	1.5917	0.215	225	382	383	383	1.000
68	1.5792	0.216	225	379	372	380	0.979
69	1.5917	0.216	225	382	382	383	0.997
70	1.5917	0.216	225	382	383	383	1.000
71	1.5917	0.214	224	382	382	383	0.997
72	1.5792	0.214	225	379	380	380	1.000
73	1.5875	0.221	225	381	379	382	0.992
74	1.5917	0.215	222	382	383	383	1.000
75	1.5792	0.215	225	379	380	380	1.000
76	1.5792	0.225	225	379	380	380	1.000
77	1.5833	0.241	225	380	381	381	1.000
78	1.5917	0.215	225	382	382	383	0.997
79	1.5875	0.221	225	381	380	382	0.995
80	1.5875	0.217	225	381	381	382	0.997
81	1.5917	0.225	220	382	380	383	0.992
82	1.5833	0.214	222	380	381	381	1.000
83	1.5917	0.215	225	382	381	383	0.995
84	1.5917	0.218	225	382	383	383	1.000
85	1.5792	0.222	225	379	378	380	0.995
86	1.5792	0.238	225	379	380	380	1.000
87	1.5792	0.229	225	379	376	380	0.989
88	1.5875	0.227	225	381	382	382	1.000
89	1.5792	0.218	225	379	380	380	1.000
90	1.5917	0.216	225	382	383	383	1.000
91	1.5792	0.216	225	379	378	380	0.995
92	1.5833	0.225	225	380	381	381	1.000
93	1.5917	0.220	220	382	382	383	0.997
94	1.5792	0.214	225	379	380	380	1.000
95	1.5792	0.238	223	379	380	380	1.000
96	1.5792	0.238	225	379	379	380	0.997
97	1.5917	0.233	221	382	383	383	1.000
98	1.5792	0.229	219	379	380	380	1.000
99	1.5917	0.218	223	382	382	383	0.997
100	1.5792	0.215	225	379	379	380	0.997
101	1.5792	0.218	225	379	380	380	1.000
102	1.5792	0.218	216	379	376	380	0.989
103	1.5792	0.229	225	379	379	380	0.997
104	1.5875	0.217	225	381	382	382	1.000

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
105	1.5917	0.229	211	382	383	383	1.000
106	1.5833	0.232	225	380	380	381	0.997
107	1.5792	0.218	225	379	379	380	0.997
108	1.5792	0.214	225	379	380	380	1.000
109	1.5917	0.220	225	382	381	383	0.995
110	1.5917	0.220	225	382	383	383	1.000
111	1.5792	0.229	225	379	380	380	1.000
112	1.5833	0.218	225	380	379	381	0.995
113	1.5792	0.248	225	379	379	380	0.997
114	1.5792	0.229	223	379	380	380	1.000
115	1.5917	0.220	224	382	383	383	1.000
116	1.5917	0.233	200	382	382	383	0.997
117	1.5792	0.222	225	379	378	380	0.995
118	1.5875	0.214	225	381	382	382	1.000
119	1.5792	0.218	224	379	380	380	1.000
120	1.5792	0.214	225	379	380	380	1.000

4 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

θ	$ D_3 $	B_{240}	rep bits	mod bits	fill	rep bits at $N=100/200$
0.000	0.447	380	381	381	1.000	159/316
0.100	0.447	380	380	381	0.997	157/318
0.200	0.447	380	377	381	0.990	159/317
0.330	0.447	380	381	381	1.000	155/318
0.500	0.453	381	379	382	0.992	159/317
0.610	0.453	381	376	382	0.984	160/318
0.750	0.453	381	382	382	1.000	160/317
0.900	0.447	380	381	381	1.000	160/318
0.057	0.447	380	381	381	1.000	159/318
0.085	0.447	380	381	381	1.000	159/318

Example Sturmian words (first 120 letters):

$\theta = 0.00$:

122121221212212122121221212122121221212122121221212
212122121212212122121221212122121221212212122121221

$\theta = 0.10$:

122121212212122121221212212122121212212122121221212
122121221212212122121221212122121221212212122121212

$\theta = 0.20$:

121221212212122121221212212122121212212122121221212
122121212212122121221212212122121212212122121221212

$\theta = 0.33$:

121221212122121221212122121212212121221212122121221
121221212212122121221212122121221212122121221212212

5 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling $2^{B_N+1} - |x|$ — the periodic ghost lines of the trajectory zoo. Throughout, $S(x) = (3x + 1)/2^{v_2(3x+1)}$ and a period- p word w with sum B satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \quad A_w = \sum_{j=0}^{p-1} 3^{p-1-j} 2^{B_j}, \quad B_j = b_1 + \cdots + b_j.$$

Negative cycles are exactly those with $2^B < 3^p$ (denominator negative).

$x = -1$: word (1), period 1

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \quad v_2 = 1, \quad S(-1) = \frac{-2}{2} = -1.$$

Cycle formula: $p = 1$, $B = 1$, $A_w = 3^0 2^0 = 1$, and $x = 1/(2^1 - 3^1) = 1/(-1) = -1$. This is the 2-adic limit of the all-ones word: $b \equiv 1$ forces $x \equiv -1 \pmod{2^k}$ for every k .

$x = -5$: word (1, 2), period 2

$$\begin{aligned} 3(-5) + 1 &= -14 = -2^1 \cdot 7, & v_2 &= 1, & S(-5) &= -7, \\ 3(-7) + 1 &= -20 = -2^2 \cdot 5, & v_2 &= 2, & S(-7) &= -5. \end{aligned}$$

Cycle formula: $p = 2$, $B = 1 + 2 = 3$, partial sums $B_0 = 0$, $B_1 = 1$:

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \quad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean $\frac{3}{2} < \log_2 3$: in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

$x = -17$: word (1, 1, 1, 2, 1, 1, 4), period 7

$$\begin{aligned} 3(-17) + 1 &= -50 = -2^1 \cdot 25, & S: -17 &\mapsto -25, \\ 3(-25) + 1 &= -74 = -2^1 \cdot 37, & -25 &\mapsto -37, \\ 3(-37) + 1 &= -110 = -2^1 \cdot 55, & -37 &\mapsto -55, \\ 3(-55) + 1 &= -164 = -2^2 \cdot 41, & -55 &\mapsto -41, \\ 3(-41) + 1 &= -122 = -2^1 \cdot 61, & -41 &\mapsto -61, \\ 3(-61) + 1 &= -182 = -2^1 \cdot 91, & -61 &\mapsto -91, \\ 3(-91) + 1 &= -272 = -2^4 \cdot 17, & -91 &\mapsto -17. \end{aligned}$$

Cycle formula: $p = 7$, $B = 1+1+1+2+1+1+4 = 11$, so $2^B = 2048 < 2187 = 3^p$; partial sums $(B_0, \dots, B_6) = (0, 1, 2, 3, 5, 6, 7)$:

$$A_w = 3^6 \cdot 1 + 3^5 \cdot 2 + 3^4 \cdot 4 + 3^3 \cdot 8 + 3^2 \cdot 32 + 3 \cdot 64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363,$$

$$x = \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \quad (139 \cdot 17 = 2363).$$

Letter mean $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$: again sub-critical, again a positive-world ascender. Note how close $11/7$ is to $\log_2 3$ — the -17 cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ($x_{\min} \leq Cp^{\tau+1}$) governs.

6 The June 2026 constraints, and the surviving candidate list

Since this catalog was first compiled, the program proved a stack of further theorems (companion notes: narrow-band exclusion; two clocks; universal ghost clock; CIC round chain; faithfulness). This section records what they jointly do to the counterexample profile, and then lists what now survives.

The new constraints, in catalog form

- *Exact narrow-band exclusion.* No acyclic orbit dwells more than $CY^{\theta(D)}$ consecutive steps in a band $[Y, 2^{2^D}Y]$; the count of confined words is an exact integer (no Perron estimates), and the counting capacity is exhausted at $D^* = 1.246$ (band ratio 5.63). Bounded-discrepancy words ($\sup |S_j| < 1.246$) are unconditionally non-realizable by divergent orbits — the Sturmian section above is now a corollary. In a ratio- ≤ 2 band the dwell bound is *logarithmic* (the dyadic strip carries one word per phase).
- *The phase diagram has two arithmetic lines.* Vertical: $D^* = 1.246$. Horizontal: the binary bias ceiling $\varepsilon_{\text{bin}} = 0.449$ — calibrated $\{1, 2\}$ -words cannot carry more order-3 shadow bias than this; exceeding it requires letters $b \geq 3$, which are descent-heavy. Saturation $\varepsilon^* \approx 0.54$.
- *Every letter is a metered ghost approach.* $b=1$ runs meter -1 (length exactly $v_2(x+1)-1$); $b=2$ runs meter the trivial cycle $+1$ (depth decrement exactly 2, parity handoff); $b \geq 3$ letters read $v_2(x + \frac{1}{3})$. Universally: every signature σ of depth K has phantom root $\rho_\sigma = C_\sigma/(2^K - 3^\ell)$ (the cycles above are instances), alignment drops exactly K per ridden block, rides last exactly $\lfloor (a-1)/K \rfloor$ blocks, and alignment a at height x costs $2^a \leq |2^K - 3^\ell|x + C_\sigma$. Amortized over *all* ghost families of depth ≥ 3 , the available drift gain is $R = 0.0882$ bits per letter against a 0.415-bit budget.
- *Supply is consumed without replacement.* An acyclic orbit visits each integer once: each dyadic band holds only 2^{k-g} points of depth g per ghost, ever. Consequences: at most 2^{k-3} upward crossings of level 2^k ; a divergent orbit spends $O(X \log X)$ lifetime steps below X .
- *The survival coin.* Parsed at 1-run starts, life against the letters- ≤ 2 regime is the reading $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$, death iff odd. On Haar fibers the coin is exactly i.i.d. fair-thirds; on ghost-aligned seeds it is deterministic (LTE): Mersenne $2^g - 1$ survives round one iff g is odd with $v_2(g-1)$ even; the 3-smooth families $2^g 3^a \mp 1$ cascade in closed form, with round-two readings such as $v_2(a' + 2 - \text{Log}_3(-5))$ — the 2-adic *logarithm of the -5 ghost above* — and every tested seed dies by round 7.
- *The $\{1, 2\}$ alphabet has an entropy deficit.* $\theta_{12}(D) \leq 0.97907 < 1$ at every band width: for an orbit with no letter ≥ 3 the phase diagram's allowed region is empty — no band of any ratio offers refuge.
- *The blind spot bites.* A positive integer has $\log_2 x_0$ bits, period: after $\approx \frac{2}{3} \log_2 x_0$ steps the word has consumed them all and every later letter is forced by the $\times 3$ carry cascade. The infinite word has Kolmogorov complexity $\leq \log_2 x_0 + O(1)$ while repetition rigidity demands near-maximal factor complexity in banded windows. Realizable words are a countable, computable family (of the 2^{30} depth-30 words over $\{1, 2\}$, exactly 66 are realized below 10^6).

The composite profile

A counterexample must therefore be: a divergent orbit, fleeing upward (escape $O(X \log X)$, single excursions below X sublinear), never pausing in any narrow band, calibrated to mean $\log_2 3$ yet carrying non-coboundary bias below the ceilings (0.449 binary, ≈ 0.54 saturated), financing every ascent through without-replacement ghost-depth pools and an amortized 0.088-bit census margin, beating an i.i.d. fair-thirds coin forever on its $+1$ -handoff parities, with a word of near-maximal local diversity generated — after a free opening of $\frac{2}{3} \log_2 x_0$ moves — by a fixed finite carry transducer with no further choices.

The candidate list, June 2026

No specific integer candidate exists: every concrete object ever computed in this program is either a 2-adic ghost or dies with a finite certificate. What survives is six structural profiles. For each we list its best known approximant and the meter that taxes it.

profile	best known approximants	taxing mechanism / status
C1. Carry-cascade pseudo-random divergent (generic profile: calibrated, biased, high-complexity)	the 120 pilot words of this catalog (all ghosts); integer side: survival champions 27, 4591, 31911, 517191, 687871 (depths 17, 27, 33, 38, 50; 59–76% of each life is in the forced carry regime)	fair-thirds survival coin; census margin $4.7\times$; all known members die; champion height grows $\approx 2^{D/2.2}$ with depth
C2. Binary violator: letters $\{1, 2\}$ forever ($\mathcal{C}_\infty \cap \mathbb{N}$, the CIC class)	none positive; the 2-adic members are the Sturmian-type ghosts of §3 and their high-complexity cousins	entropy deficit $\theta_{12} < 0.9791$: no band of any width is a refuge; parity coin must read even forever
C3. Parity-channel rider (word biased toward $(1, 1, 2)^\infty$)	integers 2-adically near $-19/11$; least representatives ride the full diagonal (ghosts)	the $K=4$ clock of $-19/11$: alignment a costs height $11x + 19 \geq 2^a$, rides last $\lfloor (a-1)/4 \rfloor$ blocks
C4. Ghost-aligned seeds (3-smooth neighborhoods $2^g 3^a \mp 1$, incl. Mersenne)	round-one survivors $2^5-1, 2^{13}-1, 2^{17}-1, \dots$ ($g-1 \equiv 2^{2t} \pmod{2^{2t+1}}$); family champion $2^{705}-1$ ($n=704$: six rounds, readings 8, 2, 2, 2, 2, 2, 3)	deterministic LTE cascade; ghost-log readings; every tested seed ($n \leq 3000$) certified dead by round 7
C5. High cycle (near-balanced periodic word, B/p a convergent of $\log_2 3$: 19/12, 65/41, 84/53, 485/306, ...)	the $-1, -5, -17$ cycles above (negative side); no positive candidate below current k -cycle exclusions	cycle formula $x = A_w/(2^B - 3^p) + \text{Baker pinning } x_{\min} \leq Cp^{\tau+1} + \text{narrow-band period localization}$
C6. Exponent-level conspirator (round-chain exponents tracking 2-adic digits of ghost logarithms, e.g. $N_0 = \text{Log}_3(-5) = 2556611305511861515 \pmod{2^{64}}$)	none known; alignment at exponent level costs height at exponent level	the lifted clock: ghosts-of-ghosts; p -adic Baker bounds the readings effectively

The list should be read as a specification of where a counterexample would have to hide, not as evidence that one exists: profiles C2–C6 are each pinned by an exact meter with a positivity cap, and C1 — the generic impostor — is the quenched calibration wall itself, the single open core to which every known framework (ours, Tao’s, the LLM-collaboration manuscript audited in the companion notes) independently reduces.

7 Interlude: could a counterexample carry a message?

A speculation, wild but within reason. The counterexample signature is near-critical drift + persistent non-coboundary bias + high local word complexity + mixed (2, 3)-adic coherence. Suppose someone wanted to *hide a message* in such an object. Could word complexity communicate information?

The channel, and its exact capacity

Any finite message can be written into an orbit’s opening. Each scripted letter costs $\approx \log_2 3 = 1.585$ bits of seed (the cylinder modulus of §2), so a seed of n bits buys a free composition of $\approx 0.63n$ letters — and then the blind spot takes over: the class pins the seed, and *every subsequent letter is forced by the $\times 3$ carry cascade*. The total information content of an entire infinite Collatz word is $K(\text{word}) \leq \log_2 x_0 + O(1)$: a counterexample can only ever say one thing — its own name. The survival champions live this: 59–76% of their lives are in the forced regime, improvisation on a theme they no longer control.

Worked example: the 56-bit integer

60822829639107739

has a Syracuse orbit whose first 40 valuations spell GHOST in ASCII ($b=1 \mapsto 0$, $b=2 \mapsto 1$), at 1.475 seed-bits per scripted letter.

The Kraft refinement: why 56 bits for a 40-bit message

The cost of a letter is not one bit but b bits: the constraint $v_2(3x+1) = b$ pins b binary digits of the seed (plus one overall). In the naive scheme a message-bit 0 costs 1 seed-bit and a message-bit 1 costs 2; GHOST's 40 bits hold 18 ones and 22 zeros, so $B+1 = 22 + 2 \cdot 18 + 1 = 59$, and the printed integer is the least member of a 2^{59} -modulus class that happened to start with three zero bits (a $\sim 1/8$ accident): 40 message bits, 59 bits of congruence conditions, 56-bit smallest solution.

The cost structure — one letter per cost class b , cost b — satisfies Kraft with *equality*, $\sum_b 2^{-b} = 1$: *the Syracuse alphabet is a complete binary prefix code*, which is the lossless-coding conjugacy of the faithfulness note in information-theoretic clothes. The optimal encoder therefore maps the letter b to the bit-block $1^{b-1}0$ (pad the message with a final 0): each letter then carries exactly as many message bits as it costs, and the rate is 1 seed-bit per message-bit. For GHOST this gives 23 letters (largest letter 5), $B+1 = 42$, and the performing integer drops to the 42-bit

3925048663003,

1.050 seed-bits per message-bit — essentially capacity (the excess is the padding zero and the +1 in the modulus). Read backwards, the same identity is why nothing beyond the seed can ever be communicated: the letter stream is a rate-1 recoding of the seed's binary digits through the carry transducer; the channel runs at capacity before you have said anything.

The information paradox

To survive, the stream must *look like* a random source of about 1.4 bits per letter (the calibrated entropy ceiling, less the Legendre tax t^* once the bias exceeds the free basin $\varepsilon_{\text{nat}} \rightarrow 0.1397$), with near-maximal factor diversity in every banded window at scale $p \approx \log_2 Y$, forever — repetition rigidity tolerates nothing less. Yet its true information rate is asymptotically *zero*: finite seed, infinite output. By the completeness theorems of the faithfulness note, the statistical tests available to the dynamics are exactly the finite-shadow algebra; so a counterexample is a pseudorandom generator engineered to *fail precisely one named test* (the chosen non-coboundary character, where it must show constant bias) *while passing all the others* — built not by a cryptographer but by carry propagation. The budgets are quantitative: bias up to 0.1397 is free; every structured ghost channel combined supplies at most $R = 0.0882$ bits per letter of the 0.415 needed ($\leq 21\%$ efficiency); the rest must come from the unstructured remainder where no finite certificate sees it.

How such a word would function

A self-feeding stream cipher. State: the binary tape of x_n . Output: the v_2 -readings at the bottom. Update: multiply by 3, propagate carries upward, truncate what was read. The opening $0.63 \log_2 x_0$ moves are the composer's; afterwards the machine's "randomness" is the binary digit stream of 3-power multiples of its own past — it reads its exhaust as fuel. By the round chain, each life-or-death reading is $v_2(3^\ell m - 1)$, so the functional requirement is that *the powers of 3, in base 2, along this one orbit, are non-normal in a precisely coordinated pattern*. That is the entire miracle, localized — the regime where mathematics has only logarithmic digit bounds (Stewart), and exactly where the program's wall stands. A counterexample would be a finite phrase whose forced elaboration is an unending, never-self-quoting, undetectably biased improvisation: a message that continues writing itself after its author has run out of ink.

Play with it

Spell any ASCII message into an orbit and read it back (from the project root `shadow_frontier/`):

```
python3 - <<'EOF'
import sys; sys.path.insert(0,'search')
from frontier import realize
```

```

def spell(msg):
    # naive: 0->b=1, 1->b=2
    bits = ''.join(f"{ord(c):08b}" for c in msg)
    word = [1 if c == '0' else 2 for c in bits]
    return realize(word)[0]

def spell_kraft(msg):
    # optimal: letter b <-> 1^(b-1) 0
    bits = ''.join(f"{ord(c):08b}" for c in msg) + '0'
    word, run = [], 0
    for c in bits:
        if c == '1': run += 1
        else: word.append(run + 1); run = 0
    return realize(word)[0]

def read_kraft(x, nchars):
    # integer -> message
    out = []
    while len(''.join(out)) < 8 * nchars:
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append('1' * (b - 1) + '0'); x = t >> b
    s = ''.join(out)[:8 * nchars]
    return ''.join(chr(int(s[i:i+8], 2)) for i in range(0, len(s), 8))

print(spell("GHOST"))          # 60822829639107739   (56 bits)
r = spell_kraft("GHOST")
print(r)                       # 3925048663003       (42 bits)
print(read_kraft(r, 5))        # GHOST
EOF

```

```

# render the four-panel candidate image of your message-orbit:
python3 search/candidate_image.py --seed 60822829639107739 \
    --steps 40 --out viz/cand_ghost_msg.png

```

Things to try: longer messages (cost ≈ 11.8 seed bits per character — `spell` of a 100-character message returns a ≈ 1180 -bit integer, instantly); watch the height panel during the scripted prefix (your bit pattern *is* the drift: letters 1 ascend, 2 descend, so the orbit's silhouette draws your message's bit-density); read past the end of your message to watch the carry cascade take over the composition — the letters after $8 \cdot |\text{msg}|$ are the automaton's, not yours, and (the $(3/4)^D$ law) a random message's orbit takes a letter $b \geq 3$ within a handful of steps of being left to itself; try to compose a message whose forced continuation survives long — that is exactly the champion-hunting problem, and the house always wins.

8 Collatz Elevator Pitch

One-liner. "We can prove the average number loses this game. We can't prove that this particular number does — and every number is a particular number."

Thirty seconds. "The rule is simple: take a number — if it's odd, triple it and add one; if it's even, halve it. The conjecture says everyone eventually crashes down to 1. Now, the ups and downs of any number look exactly like coin flips, and the coin is rigged downward — so on average, everything falls. That part is basically understood. The problem is that nothing here is actually a coin flip. Each number's entire future — every up, every down, forever — was fixed the moment you chose it, written into its digits. So to prove the conjecture, you'd have to rule out that somewhere out there is one number whose predetermined script just happens to dodge the crash forever. Probability can't do that: probability talks about most numbers, never about that one. It's the same wall as: we can prove almost every number has perfectly shuffled digits, but nobody can prove $\sqrt{2}$ does — even though $\sqrt{2}$ is sitting right there, fully determined."

The twist, if she wants one more floor. "Here's what makes it maddening — and what the last year of work actually showed. We've cornered the hypothetical escape artist. It can't follow any pattern: we proved that if its moves ever repeat, even once, in the wrong circumstances, that repetition snaps shut into a loop, and all loops are dead. So a survivor would have to move in a way that's indistinguishable from pure chance — never repeating, perfectly balanced on the knife edge between rising and falling. But it also can't actually be random, because random walkers provably crash. It would have to be a perfect impostor: a fully scripted number flawlessly impersonating chance, forever, while hiding a secret bias it needs to stay alive. We're fairly sure that impostor can't exist — being biased and looking unbiased pull against each other — but proving a specific, predetermined thing can't perfectly imitate randomness is something mathematics doesn't yet know how to do. That's the whole obstruction, in one sentence: we can beat every strategy, but we can't beat 'no strategy.'"

A note on your original phrasing: "deterministic numbers that behave totally randomly" invites the response "then they crash, like random things do —done!" The fix is to put the burden where it really sits: deterministic numbers that we can't prove aren't perfectly disguised as random. The casino version also lands well: "We've proven the house always wins on average. The conjecture asks us to prove no individual gambler beats the house forever — when every gambler's every move was decided before they walked in."

A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

```
#1 (mean 1.583,  $|D_3|=0.214$ ):
11141121112112111112114421114112222111122212112122212112221
1122211224112121222212121211111121221222111411112211121
2221111111124222221111112121141121211121422111414212111141
21121141222121111111222212114121211221411121111114422241

#2 (mean 1.579,  $|D_3|=0.214$ ):
222222222222211111121121142111121142111122122221114241211
1111124121112114221111411121111221122122121111112121111211
212221112121111422142411121112141121114421211114121114221
21121111111122121111212112211212111411211412422212222111

#3 (mean 1.579,  $|D_3|=0.214$ ):
222112111112112421221112111211112121221112111211221144412
42222211122121211111112111111211121141212111222422214111212111
11241111211222421112111211122121122111124112111224111121221
22412211411211112212241211242112221211144112111211212111122

#4 (mean 1.583,  $|D_3|=0.239$ ):
1111111111111111111111111111111111112221141142141221114212
11142114111221421242112211441122411112111221212211222221111
11111222122121111111111121421121124112212121142244122121211
121411412211121111211221111422424421212112121142112221121111

#5 (mean 1.592,  $|D_3|=0.220$ ):
1111111111111111111111111111121121111412414211121212411112242
12112112112121111421114122211124212211121122211111111122111
2212221124121211114224121122224421212141111124111221221212
2121121141111112211214211112221112222414212211221214211224

#6 (mean 1.579,  $|D_3|=0.215$ ):
22222224222224211211114124212114112212222111112112241221
11112112111111111122111122222421114212214211111112422221
111112221111121112221411112222112121124211111141112121222
1111121111121111221122121421114112212114121411121211212112

#7 (mean 1.592,  $|D_3|=0.215$ ):
121112111211212111122412111212141121211411241122124222211
22122111111221111122121222421411121112212111222111211112112
212422141111221412421111222214112142112111212111112211111
11114121222242111111221211221111212421212114112211222141211

#8 (mean 1.579,  $|D_3|=0.216$ ):
12211111111212121111141121222111112111121111211142211414
2222211412111114212221212111111211412212121112212121421121
41212112411221111242211222111111411421211122222411112111221
21412212111411211121122114122411111211211241212122121122221

#9 (mean 1.579,  $|D_3|=0.229$ ):
222222222222422222222222222222222221122211221144111
12112112211122111221111121114212411112221112112111122241111
2412121211411111211112111221111122111211114121212114211111
1111221122111111211111422421211211111221111111111112112421
```

#10 (mean 1.579, $|D_3|=0.233$):
111112111222121222121211114111414222441214211212124111122211
111111111121121421111411122221211224211422141122144142111121
1421411112122111112122224111111111111112242212121221111111
11122142211212111111111422112111121211121112111412221

#11 (mean 1.579, $|D_3|=0.229$):
2222242222224222222112211221241221221112211142211112122112
1111121112121211112111221121211221122111122241121211241141
12121111421112112112211141212121121422211112112411211222121
2242212122122121121114212111214122111111221111111212211111

#12 (mean 1.579, $|D_3|=0.216$):
422211212114121122244111122221112411221221112121112441124421
11222211111211121112212111221221211111111121242212112122
114122214121111112412411112141121211111141112221141211212
2111121112111111122242112111421111121221212121111212211221

#13 (mean 1.592, $|D_3|=0.233$):
11222212111121
12212421411221111112211111222122141114421122211212442114111
121241421122224122111422121211222421122114421142422121221111
211124144212122112121222112111241112111211121114212122141

#14 (mean 1.579, $|D_3|=0.215$):
2422222224222222211111111212211111111111111221121224211
2111212122221121221214121121221121221212121211121142111121
21211122111411112112124412221121112211112212412411211222212
2112111111121212141141111212411122414112412212121111211411

#15 (mean 1.592, $|D_3|=0.216$):
112441122111211112112211221
12122121211412412142112412211111122412112222211111222211122
122212221421212121422224112141112121112221411414111122111
21121122112124211211412421121122411421222221114112222111111

#16 (mean 1.583, $|D_3|=0.214$):
2122141111121112121111212211121211112122414224112111221122
2411221421212111112412112444112212211111122212124222211214
2111214211111121221111242112212141211111221122111111121222
114121211411211111241222121221211112121212212121112111122

#17 (mean 1.592, $|D_3|=0.215$):
111211212242121222211221222111112122421112112112222114111212
21111124212241211212121112211141122141121124221112221211411
4111114212112211121122111221112111111221221121221121114112
2122211111111411141121121411111222112124211121122242421222

#18 (mean 1.579, $|D_3|=0.218$):
22222224242211124111122122121141122122112121221122114241112
1122421122111212111411112121141111222121121212412111121222
12114121111222221112111122114124141122111221111212212221111
212214221121121111212121111222111421121112211112111122111

#19 (mean 1.592, $|D_3|=0.220$):
112411111211112121112411212111212211211111412211111211112221
12121211121221211212111411111112141122122442121122112222112
2114142111411121122122441111121121112112221412411141212222
11112112212441121211112212124112124112124242212112211111111

#20 (mean 1.579, $|D_3|=0.260$):
22222422222222222222222222222222222122111112211111221124112
21111211221122412111411112221121221211222111111412122212212
42111421212422121211211221111112112211112211122124221111211
121211111121222121111211111111111114112111241121222122212111

#21 (mean 1.579, $|D_3|=0.272$):
22222222222222222222222222222222222121211112212211121122222121
1111112212121111222111121111211112221112122111141121414121
1112112111214222121212121211224212122111124142121111211212121
1211211121111112212121122112112221222124412212111111412122

#22 (mean 1.592, $|D_3|=0.220$):
11111111111111111111111111111211112242124211142124122111112241
221114214212211122221221414421121112221121141222111211112212
1141111211141121211112111122441121211221121222421112111114
12111121211212124242212121241212211122411111212122122211212

#23 (mean 1.587, $|D_3|=0.228$):
142222112411411112111121121122414422122111111211211222412
222121111211221211111211122114112121212141111121111221122122
12422121211111141214211211222111121211114121112421211112112
122411111111111112111122222111122424411124114214111122111

#24 (mean 1.587, $|D_3|=0.246$):
1122122211412
1212121224142114111111121421221141112112421212222414114212

[illegible]

#39 (mean 1.587, $|D_3|=0.221$):
224111212221411111122412221141121111111411212111241211211
11121112222121411114221141122112211221211121221211112111111
241411211121242121211142222111141211211211112121212111111
1442212111121241221221112221121121114221212111122444212211

#40 (mean 1.579, $|D_3|=0.218$):
222222222214112211111121411121111211112111122241122111
112121411111224111211112222122112211122112112212111412142212
222122122221222112111121114112221121211112141111221122211411
1211112112124114122122122121211411221441211111111242111212

#41 (mean 1.592, $|D_3|=0.225$):
111111111111111111111111111111111112211211111122141221412211
21111222111214122112211112221111112211121121421241424241211
422121111141111141111241112121222111222122214122221221212
24212122411142144221211112221112124111141212242111112412212

#42 (mean 1.579, $|D_3|=0.238$):
2222222222222222222222222211221114112111111141124121111
1212411124111221121211111112111211112221111211221212121
2111141211111221111221221212121112121112121111121221111
2122411124111221222122111221112422211411212121224222212441

#43 (mean 1.579, $|D_3|=0.215$):
222422242221121224211214141211211124411212111112111124111222
111112221111222412211111111121212142112111212112224214221
22111121211112212211212111121111212211112221221121111122214
21224121121111221212111121214221111114211111212221412121

#44 (mean 1.592, $|D_3|=0.215$):
1111111111111111121112212211221111142111221111224211121211411
422211112211111222442111112211112122412111212221412211422211
1122142112211112111122122411212122112112212112121111211121214
11211211112141111222211412222112121221224411442221222112112

#45 (mean 1.587, $|D_3|=0.217$):
12112112242122111414111111111212122411221111111121111211112111
122112222211211212224111241141222241242141211221121221421412
11111112221221211142211112122111211214121111222222211412
11111112212121111121214111241111411121222112142211111141

#46 (mean 1.579, $|D_3|=0.215$):
22224222222222222222222121111121111211214221114411221111112
124111111112121111121112222111121221121111221111224122114
211121144411111421221112222111121111212212112111222211111
21411214124121411211421111212121121112114211211112111121212

#47 (mean 1.579, $|D_3|=0.229$):
2242222222222222222242224422222212211112411114411421111111
111121211211111241412111112122121211112221121211211222111211
11141112121112211111122121222211221121211211111111212211111
1214211211222122421112221222212122111111111121112121121411

#48 (mean 1.579, $|D_3|=0.215$):
242222224212211121111111121211124411411111122422121421112
211121222212212411121122122122112121221212421111141111211
1214141111111112111411211212222142121112122112122212111221
2424212212111111111112222111122211211141121111211121212112

#49 (mean 1.592, $|D_3|=0.215$):
111111111141121142222122114411211221421214242121121244211111
2122211211211212214212211411212111212212121122222121211211
2211111211221111111211212411121112211111221212111212111122
222211121141212222114421421222111112411112222412121111111

#50 (mean 1.592, $|D_3|=0.216$):
122112121421111242121122112121211221212221421421111121111121
14122111114142111122212212141212211111441121111111111121221
211144121112111211212122112211222212141212111111121114111212
1214112421242122121221221241122111141211121122421111221211

#51 (mean 1.579, $|D_3|=0.229$):
22222222122124211122412221122112111221122214122111112112
21112441121112142111211141111221111222442111111121122211121
21212212111112222211122121141221222112122111211121122112221
1111121222214111411411212111121411222121121221122111212112

#52 (mean 1.579, $|D_3|=0.248$):
2224222222222222222222222222222222222122111211122111112121
2111222111111121112411111222212111111122211121121122111211
24211111112211212242122211244211241412111122111221112111111
4111122122111212111112411221121212114411211121121121221121

#53 (mean 1.592, $|D_3|=0.218$):
111111111111111114212111442211121112121121142122141111121111
4121221214222222212111121211212222214111122121211141122212

214411114241111121422111221212111212111212121241111222111212
11121221211222112122422212122114111211122122112221222111211

#54 (mean 1.579, $|D_3|=0.215$):
2422222244222222442222222224211111212221111112111111111142
21121111212221112121111421122214111112122122112111122111
2112421112212111111212111421211111421212121111211111221122
111144221221211142211112121112221111122112111421111211421

#55 (mean 1.592, $|D_3|=0.233$):
11111111111111211441111414121422112121112211112211121111
22112121111122212111112211211214211122412111122212211141112
211112142112121241112214224112411221241211112112111212112
2211411112422412121412111141121142422212112211411111211121

#56 (mean 1.592, $|D_3|=0.279$):
1114111242124112
2411221111112112114211111412122422111122121244211112111141
44212222141112112112211214211242241114211214211212411112411
211111222121221221412111122121441221121221144141111211142

#57 (mean 1.579, $|D_3|=0.215$):
22222222211214112211211221111111211141122212224211211121111
1112114211112222112124122212214112112124111112111211241222
2141221121121411111111214211242111122121411121222141211211
1221121122142212111114222212421122112112111211121211111111

#58 (mean 1.579, $|D_3|=0.218$):
22222222222211111212111444121114121122121211112112211241244
12211141111211211211111111212221214212211112212111441112
2211211122111242211212241222121111111421211124211124212112
121121221211121111221212112221122112212111212122112221111

#59 (mean 1.579, $|D_3|=0.238$):
22422221122111112211
11211212211211214441121121111412124111211221211121121212111
112421114211422211211122122111112112411121211412111111121212
11111212211111211114211112211121112111111421121112121211

#60 (mean 1.579, $|D_3|=0.214$):
22242222222222211221112421112111241212211142111211211141221
111112112121411221111111122412111421214114111122111141112
111211212211414112111441221212111212112211124212221111122221
11124211211211112121111211412111112211112221122122221211

#61 (mean 1.579, $|D_3|=0.238$):
2222222222222224221111111121141111212112221114242121111111
12112211212421141422111122222111221111111121221111112112221
2122214212111122122211221221111211111112214221112211112222
212122221211211121142212212144122111211211122111421121212112

#62 (mean 1.583, $|D_3|=0.230$):
1111111111111111111111111111124122111112114212112242141411212
111221222122141111121111422211412114111411212111121211212241
112211412411112111224211412121121212224412242111141212111411
12212211111221124112112111122111222121121211212212221114122

#63 (mean 1.579, $|D_3|=0.222$):
22222222222222222222211412141121122211221111211112241112211
111221111121212214142211122111411211241112112211112142112211
2121142141111112121111211114111421422221112111121221241111
22212241111211211221111111112212111211122212211222121112211

#64 (mean 1.579, $|D_3|=0.215$):
2222222222222222222211111141111214121141111121212241211112
211121212111121421111114122212212111114411111114112111242112
211111412211221141122211121121411142112112112211111221212211
2212121211211112111114112422112211212211121422111212221211

#65 (mean 1.579, $|D_3|=0.222$):
22221222112121111141211214111122212111111212112122122122211
1112424112421211122111212111111221112112122222112111212211
11211212211221221221111122112122221411211121211412212121211
1421214211141122122222212111241121121111414221121111441221

#66 (mean 1.592, $|D_3|=0.216$):
11121212122111111
22112221241224211221112111211111221222411142221212112224211
12411241121144112112111222421412114111122112222212121212122
222211212114241212442421121112121421211124211242221212112211

#67 (mean 1.592, $|D_3|=0.215$):
11111212111411111222212121114111411121111121212411412221
1222221112122141114112212122221211211411212112411211122112212
2141112141112212211121411121111212112122122412141122211121
412111144211212111212412211212122212111212122211222111211222

#68 (mean 1.579, $|D_3|=0.216$):
2222222222224242424111212111121241221144212112114111111221
21211212111111124421211221222221121211121121211211141211112
112121114122121111412111211111122122112211111121241212
122121111141111111441122111111122421211212411122211112111

#69 (mean 1.592, $|D_3|=0.216$):
1111112122411112212111141111122121121224112221222221142111
111114221211242212141122224242211214242111221111221211222111
141111121212122112211111212211122114111112422121112111111
12124112221121411111411122421121112111412121121111421212

#70 (mean 1.592, $|D_3|=0.216$):
211142121122412224111224112112121121121211111412112222
11122111212141212212121111112241121211212111122111214224
1122211212121211422112221121121141114222112224111111111
124122121114121111221212211211121442111114422412111111112

#71 (mean 1.592, $|D_3|=0.214$):
111111111111112211214121421112122112211114122144221122222
1144144124242112221411121121212111112114212221112221212211
122224211221211121222212411111411212211121121112121111222
2112122112142121211121112211212412112111221412121112111112

#72 (mean 1.579, $|D_3|=0.214$):
2422422221221111141112111241121121212211221111221212121
2244111142211221112241111112121211112121211421212141242111
2121112112221122214122411112111112122142212121122111122121
122212212121212111121112141121111211122411111112114111114

#73 (mean 1.587, $|D_3|=0.221$):
12114111112241112112112422112121211411222111121111124214211
1222211111112212111212141211144111221211112121111221212121
1122111111124124212112221211114121124111111121112112121241
21112224221411112241121121112122214121212241111221111411241

#74 (mean 1.592, $|D_3|=0.215$):
111111111111111112122112121422222114111122121212121122111
111221221141242112222111122122121211212224121122221142212212
2122212211121211111222421221111111212121212111112114244142
221411211124111211411112111112211114122422122221411211124

#75 (mean 1.579, $|D_3|=0.215$):
22112411142121212121112222111111242121212121121141221211
2121114121141111212111121141121241211241111211221221224412
2211112222111121221221111121212122221141112111422221111211
1142112211211212122121112122124124112121112122112111114121

#76 (mean 1.579, $|D_3|=0.225$):
2222242111221411122111124211111121112112124111111214111112
421111211112121211111224121112142121211121111112221122122
2111221212212111411211124221242111241122112111111111212414
21114121211211112111121222121414221111211221114412214421112

#77 (mean 1.583, $|D_3|=0.241$):
111111122221112111112112111222121141412224214222221242142
2211221141242212121211121112412111212111211211112122221
12112211211422221114222211112411112121112211111221221122
12122122112421211121211221221112212211122121111221212112

#78 (mean 1.592, $|D_3|=0.215$):
1111111111121121112122211442211222212222211112112212111412
11222112221211424141111222121111211141112124411111212222111
2212221211212111211222144122111111141212222111112122221222
111122412111221112111211112111142212221421411122111114411211

#79 (mean 1.587, $|D_3|=0.221$):
14121122111111142122121211121211222211211111111141111111124
11211414222122111112112214212222121112112211111114414124222
111111241121411121112211111124211412111142111221211122112112
1241121411122211122121221121112411411121221221121111222242

#80 (mean 1.587, $|D_3|=0.217$):
111112144211112211141212112111211222111122121211121221111121
21122111112141421221211142214221211121121112112211212122142
1211122121212222114122221212111211111221241111114211122242
12211224222111241114212121211211242421212111221112111121112

#81 (mean 1.592, $|D_3|=0.225$):
111111111111111111124141111222211214121212111412114211421
1122121111112112112124121111122424111111221142112111212141
22221121112114221112224111242221111114122112122124122211112
111112121112221211224114211124221242122214211111121412211121

#82 (mean 1.583, $|D_3|=0.214$):
212112124241211121411212112122112112121212422111111121214
224111211121124412221122112221121211221141142214112424421

12421122111211111211222411121121212111221112111112112212121
1111111122122111214112211121111222111122112241211221221111

#83 (mean 1.592, $|D_3|=0.215$):
112212111124212112211211112112111141122121212221111111122
2211112241112222122212212111121222212112111422421441221212
24111222122121112141412244222111224111121121212211112111421
41111212121211414111122121221411211212111112221112121111

#84 (mean 1.592, $|D_3|=0.218$):
122111111121141121122212142221142212112221111212122121211211
121211112112122421411111141121242211112122221111211144112242
1211111111122121122122221221144122111112122212211411111214
21112121422114211121211121112114112111122212112222221112221

#85 (mean 1.579, $|D_3|=0.222$):
2121224411222214222122121211112111121114141141211111122
12212111112112122121121121121221211412221211211121122221211
22212421221121112112112412121411112212411112211221211112212
44122122121211112212122411121212122121222111211412121211111

#86 (mean 1.579, $|D_3|=0.238$):
22222222242222422222222222211221111212241212211121242112214
112111111111112112214211111221112111112121121412221111142
211221111112211112111212111214122121212111122121221122112
2121222211141142421221121211211112222212121121114111112112

#87 (mean 1.579, $|D_3|=0.229$):
422222222222222422221422221112221212111211211122111412
11122111211111422111221111121124121111221224111111121111
112122211111111111114111122411121221211111122121124222
22112222111112111211241421121221122122242211112121212211221

#88 (mean 1.587, $|D_3|=0.227$):
24214121212121214121114111222121211111212211212121212222
121112221111211111212412111211221221111111211112112122121
1212421224122121112121222111112212122221212421212121221122
11224211112222121244222111121414121111121142211111124122

#89 (mean 1.579, $|D_3|=0.218$):
4222222222141112112211121212122111111212111111122122121
12112221121212212221112412112111411211421112112111121114
241221111122111411112241411212111211112422212222221212242
2112211112112112211111122211211112122411142141124122221112

#90 (mean 1.592, $|D_3|=0.216$):
11111112112121114212212121112122112211121211112142211142111
2111221111244121112222142121111111212112112114221112421112
11112121221222121224142112211121212121212111211141421211
11112114111221212111122221221411411112244142112111241224111

#91 (mean 1.579, $|D_3|=0.216$):
42222422221214111212121212141112111141112121112111121112
1121121212212411211121412212121111111112112224111211122212
1244122141421144121211211111241221112111122221221212111211
2212211111241221111111211122112111112212114121114214121222

#92 (mean 1.583, $|D_3|=0.225$):
1111112122221122122114112112211224212222111124221211112112
124111222411211241121122112111111121224122422111111112212
2111211122121211121111221211121212224222222211122121212
11211211222124124242114121111211111221111121242111241211221

#93 (mean 1.592, $|D_3|=0.220$):
1111111111111111111122124212111114141212221211112112212111
1122141111212211111122212412112111124121121212212122112122
44242112211142112411212111122112111211211411124411411124122
22111211121122212212121142421224212411212111411122112122111

#94 (mean 1.579, $|D_3|=0.214$):
442222222222224212111111111121112211121124142112211211
11211212212122421212241121111221112212421211211141112111211
141242112111121221111222411412111112412211111111114211112
22211122111112121112142111212221221112112111121412212112211

#95 (mean 1.579, $|D_3|=0.238$):
222222222222422222242222222222222221211111111112212121
24111111211111211212211124112112111112142121121121211121221
11124122221111411122122111114222412112111121212111111112
21111221222111141112212221111242224111111111122112211111

#96 (mean 1.579, $|D_3|=0.238$):
2222222422222222221211224211112111121211212121211122122
2211222212111121114122111111121411221112211212122111122112
1211141221141111221112122111122211212414121121212422121141
122212111111211241221111122221212111111221121112421122121

[illegible]

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1224141111111221124441212212121111211224212222112122112221
1121212211122112121121212411212112211121242111211221422421

#112 (mean 1.583,  $|D_3|=0.218$ ):
1111111111121211121141221221412211122211111211141111122111
241121421211112121212411422112422111121112222144141214222121
11212112122211122211211121121111112122111112112111121121121
212442111411122212122112221122211112442422111211144221211111

#113 (mean 1.579,  $|D_3|=0.248$ ):
222242222222222222112122211122221112111142211121212111241
1111122121212122421111211222111221121122121211121122111111
1122221122111121212221121111422114212122112141411112211112
2124112112111121122112112214122221121422212121422211111112

#114 (mean 1.579,  $|D_3|=0.229$ ):
222222222222222211111211212111121212124111122411112222
1122222122111112111121121211141122211412411111222112411211
21121421211111111114412112221221111212114212411212111221222
1422121211222212122211212112121114111211211211112214211

#115 (mean 1.592,  $|D_3|=0.220$ ):
1111111111111111121121121114222221221222242212211211114111
22221212111211111111144111422121212141112422141242112112112
12121141112122122211111241214422111222122121114121121241211
12221144111211212111211121111211142111122241142221211111

#116 (mean 1.592,  $|D_3|=0.233$ ):
11111111111111111111111111111111111111111111112112112114212221244
112111422124111121241222121142222121242122112121222111222212
2214144121141111112121242114121421412124212211111211112222
11211111112114121121112411111224222141121222124124111121111

#117 (mean 1.579,  $|D_3|=0.222$ ):
2222222112222111141111122214211221212422112211111122241221
21121121121122211121112121111211112211222122421111121211422
1111111211112122224111112112112211111211121421111221121222
41222211221214111112141222221111212114124211241222114112221

#118 (mean 1.587,  $|D_3|=0.214$ ):
1241222422214111212111111212121121221111212111211121422111
21122111112111112211411112221111222121111211121111242212111
11122112421112211114112111212211112142112242211122221224144
124221222211111421211112111124241121222211111121121221142

#119 (mean 1.579,  $|D_3|=0.218$ ):
22224222222222222222222222222242112112112141111111221142
11121211121112121222122112111221211122111211211111121124111
111211112211142111111211111241421224221411112222121111122
1411211221111111211211412112211111121121111141422241211

#120 (mean 1.579,  $|D_3|=0.214$ ):
22222422224222222122411212122111121111111121211211212111
212211121121112111111211412211221112112211122241411112111
1211221212111221141211212114122111114111221211111211422444
41121111112121122241141221241241112121121211111121121211221

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